

Lab Experiment – 3

Flow Control Data Link Layer

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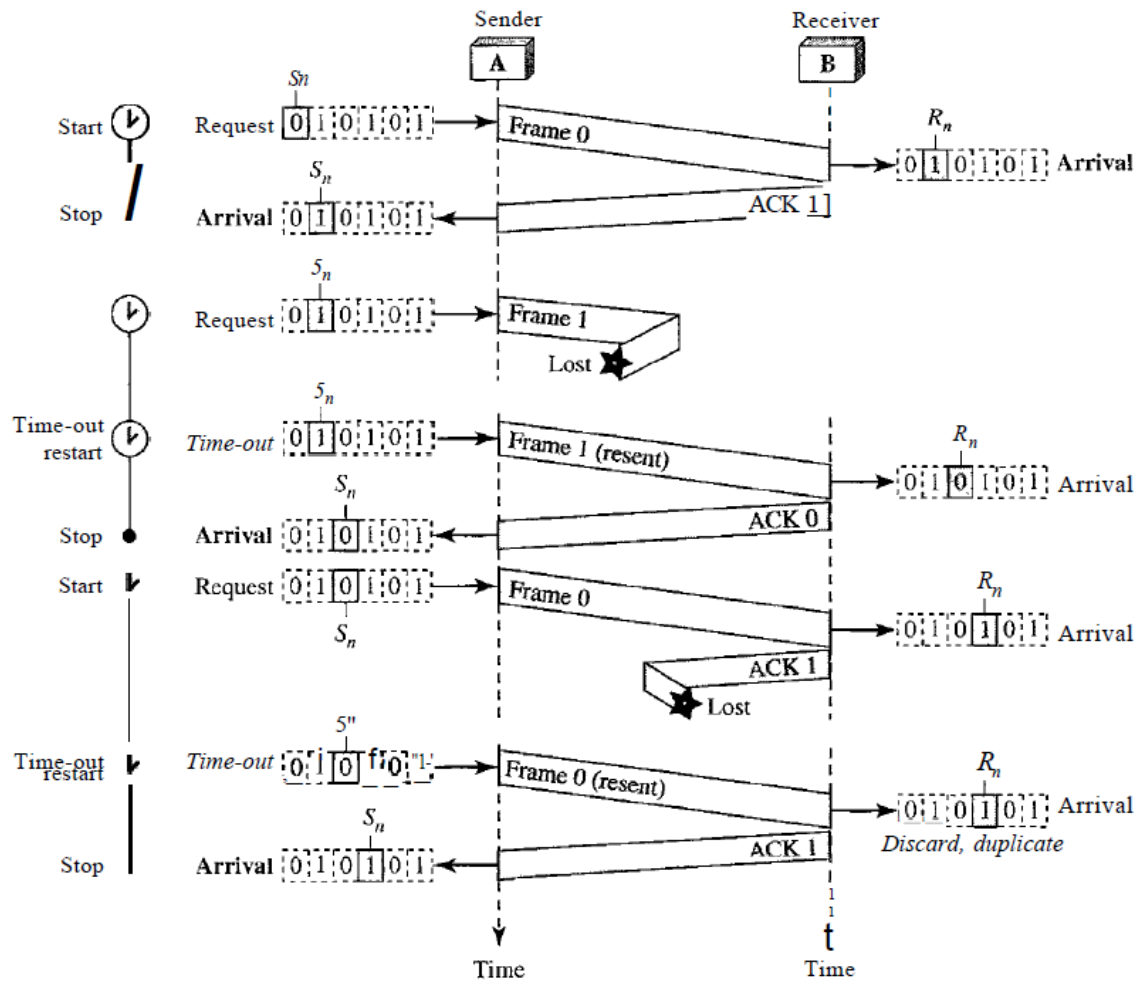
1. Flow Control – Stop and Wait Protocol –Noisy Channel

Implement Stop and Wait for Noiseless Channel – Algorithm. Create Sender and Receiver module for the implementation.

Noisy Channel – Algorithm

Algorithm 11.5 Sender-site algorithm for Stop-and-Wait ARQ (continued)

```
6  if(Event (RequestToSend) AND canSend)
7  {
8      GetData () ;
9      MakeFrame (Sn) ;           //The seqNo is Sn
10     StoreFrame(Sn);           //Keep copy
11     SendFrame(Sn) ;
12     StartTimerO;
13     Sn = Sn + 1;
14     canSend = false;
15 }
16 WaitForEvent();               // Sleep
17 if(Event (ArrivalNotification) // An ACK has arrived
18 {
19     ReceiveFrame(ackNo);       //Receive the ACE fram
20     if(not corrupted AND ackNo == Sn) //Valid ACK
21     {
22         Stoptimer{};
23         PurgeFrame(Sn_1);       //Copy is not needed
24         canSend = true;
25     }
26 }
27
28 if(Event (TimeOut)             // The timer expired
29 {
30     StartTimer();
31     ResendFrame(Sn_1);         //Resend a copy check
32 }
33 }
```



Sample Scenario

2. Implement Flow Control – Go back N ARQ with no packet loss, packet loss and Acknowledge loss

Algorithm

Algorithm 11.7 Go-Back-N sender algorithm (continued)

```
21  if(Event(ArrivalNotification)) //ACK arrives
22  {
23      Receive(ACK);
24      if(corrupted(ACK))
25          Sleep();
26      if((ackNo>Sf)&&(ackNo<=Sn)) //If a valid ACK
27      While(Sf <= ackNo)
28      {
29          PurgeFrame(Sf);
30          Sf = Sf + 1;
31      }
32      StopTimer();
33  }
34
35  if(Event(TimeOut)) //The timer expires
36  {
37      StartTimer();
38      Temp = Sf;
39      while(Temp < Sn);
40      {
41          SendFrame(Sf);
42          Sf = Sf + 1;
43      }
44  }
45 }
```

11.48

Sample Output

```
Enter the Total number of frames : 12
Enter the Window Size : 4
Sending Frame 1...
Sending Frame 2...
Sending Frame 3...
Sending Frame 4...
Timeout!! Frame Number : 1 Not Received
Retransmitting Window...

Sending Frame 1...
Sending Frame 2...
Sending Frame 3...
Sending Frame 4...
Acknowledgment for Frame 1...
Timeout!! Frame Number : 2 Not Received
Retransmitting Window...
```

Sending Frame 2...
Sending Frame 3...
Sending Frame 4...
Sending Frame 5...
Timeout!! Frame Number : 2 Not Received
Retransmitting Window...

Sending Frame 2...
Sending Frame 3...
Sending Frame 4...
Sending Frame 5...
Acknowledgment for Frame 2...
Acknowledgment for Frame 3...
Acknowledgment for Frame 4...
Timeout!! Frame Number : 5 Not Received
Retransmitting Window...

Sending Frame 5...
Sending Frame 6...
Sending Frame 7...
Sending Frame 8...
Timeout!! Frame Number : 5 Not Received
Retransmitting Window...

Sending Frame 5...
Sending Frame 6...
Sending Frame 7...
Sending Frame 8...
Acknowledgment for Frame 5...
Timeout!! Frame Number : 6 Not Received
Retransmitting Window...

Sending Frame 6...
Sending Frame 7...
Sending Frame 8...
Sending Frame 9...
Acknowledgment for Frame 6...
Timeout!! Frame Number : 7 Not Received
Retransmitting Window...

Sending Frame 7...
Sending Frame 8...
Sending Frame 9...
Sending Frame 10...
Acknowledgment for Frame 7...
Acknowledgment for Frame 8...
Acknowledgment for Frame 9...
Acknowledgment for Frame 10...

Sending Frame 11...
Sending Frame 12...
Timeout!! Frame Number : 11 Not Received
Retransmitting Window...

Sending Frame 11...
Sending Frame 12...
Timeout!! Frame Number : 11 Not Received
Retransmitting Window...

Sending Frame 11...
Sending Frame 12...
Acknowledgment for Frame 11...
Acknowledgment for Frame 12...

Total number of frames transmitted(sent + resent) are : 38

3. Implement Flow Control – Selective Repeat ARQ with no packet loss, packet loss and Acknowledge loss

Algorithm

Algorithm 11.9 Sender-site Selective Repeat algorithm (continued)

```

20  if(Event(ArrivalNotification)) //ACK arrives
21  {
22      Receive(frame);           //Receive ACK or NAK
23      if(corrupted(frame))
24          Sleep();
25      if (FrameType == NAK)
26          if (nakNo between  $S_f$  and  $S_n$ )
27          {
28              resend(nakNo);
29              StartTimer(nakNo);
30          }
31      if (FrameType == ACK)
32          if (ackNo between  $S_f$  and  $S_n$ )
33          {
34              while( $s_f < \text{ackNo}$ )
35              {
36                  Purge( $s_f$ );
37                  StopTimer( $s_f$ );
38                   $S_f = S_f + 1$ ;
39              }
40          }
41  }

```

11.62

(continued)

Output Scenario

